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Section : B

1A.

```
red@red-ubuntu-24-server:~/PES2UG24CS068-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

1B.

```
red@red-ubuntu-24-server:~/PES2UG24CS068-pes-vcs$ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
```

2A.

```
red@red-ubuntu-24-server:~/PES2UG24CS068-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

2B.

```
red@red-ubuntu-24-server:~/PES2UG24CS068-pes-vcs$ xxd .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
00000000: 626c 6f62 2031 3600 4865 6c6c 6f2c 2050  blob 16.Hello, P
00000010: 4553 2d56 4353 210a                      ES-VCS!.
```

3A.

```
red@red-ubuntu-24-server:~/PES2UG24CS068-pes-vcs$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  pes.h
  untracked:  README.md
  untracked:  index.h
  untracked:  index.c
  untracked:  commit.h
  untracked:  Makefile
  untracked:  pes.c
  untracked:  tree.h
  untracked:  commit.c
  untracked:  test_objects
  untracked:  test_sequence.sh
  untracked:  .gitignore
  untracked:  tree.c
  untracked:  object.h
  untracked:  test_tree.c
  untracked:  test_objects.c
  untracked:  object.c
```

3B.

```
red@red-ubuntu-24-server:~/PES2UG24CS068-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776451887 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776451887 6 file2.txt
red@red-ubuntu-24-server:~/PES2UG24CS068-pes-vcs$
```

4A.

```
red@red-ubuntu-24-server:~/PES2UG24CS068-pes-vcs$ ./pes log
commit 5e1680546c6118c18d4533461f2afd7bce2959a0022eb98b18c35ad94ee4b756
Author: Anishwar Rajaram <PES2UG24CS068>
Date:   1776451971

    Add farewellt_id!

commit 10506d6da3a29b5efea7d14d7db5edd94c1ac4dfb77590cbf554f0a6d071569c
Author: Anishwar Rajaram <PES2UG24CS068>
Date:   1776451971

    Add worldellt_id!

commit ffb8fcce58d56af871341979f38af236b02b476fda2ba7ea47fdf43c94ea071f
Author: Anishwar Rajaram <PES2UG24CS068>
Date:   1776451971

    Initial committial commiq
```

4B.

```
red@red-ubuntu-24-server:~/PES2UG24CS068-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/10/506d6da3a29b5efea7d14d7db5edd94c1ac4dfb77590cbf554f0a6d071569c
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/5e/1680546c6118c18d4533461f2afd7bce2959a0022eb98b18c35ad94ee4b756
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/85/e367b76c2417b47a95f46208835ad8700dd58335171c94a497974be6191900
.pes/objects/96/c0d5c28dd3b719606c36723cd16aefbc73b3dc8b74de61070183357d6968b9
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/f1/0239b620372163c2dd06ac33189180c55edf1c56b15990a1cf44e00b1b0e59
.pes/objects/ff/b8fcce58d56af871341979f38af236b02b476fda2ba7ea47fdf43c94ea071f
.pes/refs/heads/main
```

4C.

```
red@red-ubuntu-24-server:~/PES2UG24CS068-pes-vcs$ cat .pes/refs/heads/main
5e1680546c6118c18d4533461f2afd7bce2959a0022eb98b18c35ad94ee4b756
red@red-ubuntu-24-server:~/PES2UG24CS068-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
```

Q5.1
A complete replacing of the working directory would be warranted to change HEAD.
This involves reading the commit object to get its root tree hash, recursively traversing the tree to write all blobs as actual files in the working directory, and deleting any files that exist in the current working directory but are not present in the target tree.

Q5.2
We would check the current commit's version and the target commit's version. If the hashes of the current file status and it's stored version differ, and the target branch also has a different hash, then we must not allow the checkout.

Q5.3
They become orphaned commits with no branch reference pointing to them.
They exist in the object store but are not reachable from any branch name, making them invisible in normal logs and vulnerable to garbage collection

Q6.1
We start from all branch references and recursively walk through every commit, tree, and blob, recording each object's hash in a lookup set. Then scan every file in the object store and delete any whose hash is absent from that set.
I would use a B tree

Q6.2

If garbage collection deletes an unreachable blob after a commit has started using it but before the commit updates the branch pointer, the commit ends up referencing a deleted object, corrupting the repository. Git avoids this by using atomic reference updates and a prune expiration window (default 2 weeks), ensuring no recently created objects are ever deleted.